

1 Solution — GIAM 4.1

Problem 5

1.1 Problem

What, in Set theory, corresponds to the notion t (a tautology) in Logic?

1.2 Answer

The **universal set** U — the whole universe of discourse.

1.3 Why

This is the mirror image of the previous exercise, read through the same bridge. For an open sentence $P(x)$, the truth set is

$$S_P = \{x : P(x)\},$$

so the question is: *which sentence has everything in its truth set?*

*$S_P = U$ means every x in the universe satisfies $P(x)$ — that is, $P(x)$ is **true** for every x .*

A sentence true no matter what is precisely a tautology, denoted t in the text (as in $A \vee \neg A \cong t$).

Checked on the universe $\{1, \dots, 10\}$: the sentences “ $x = x$ ”, “ x is even or x is odd” and “ $x < 100$ ” all have truth set equal to all ten elements, while “ x is even” captures only five.

1.4 The Picture

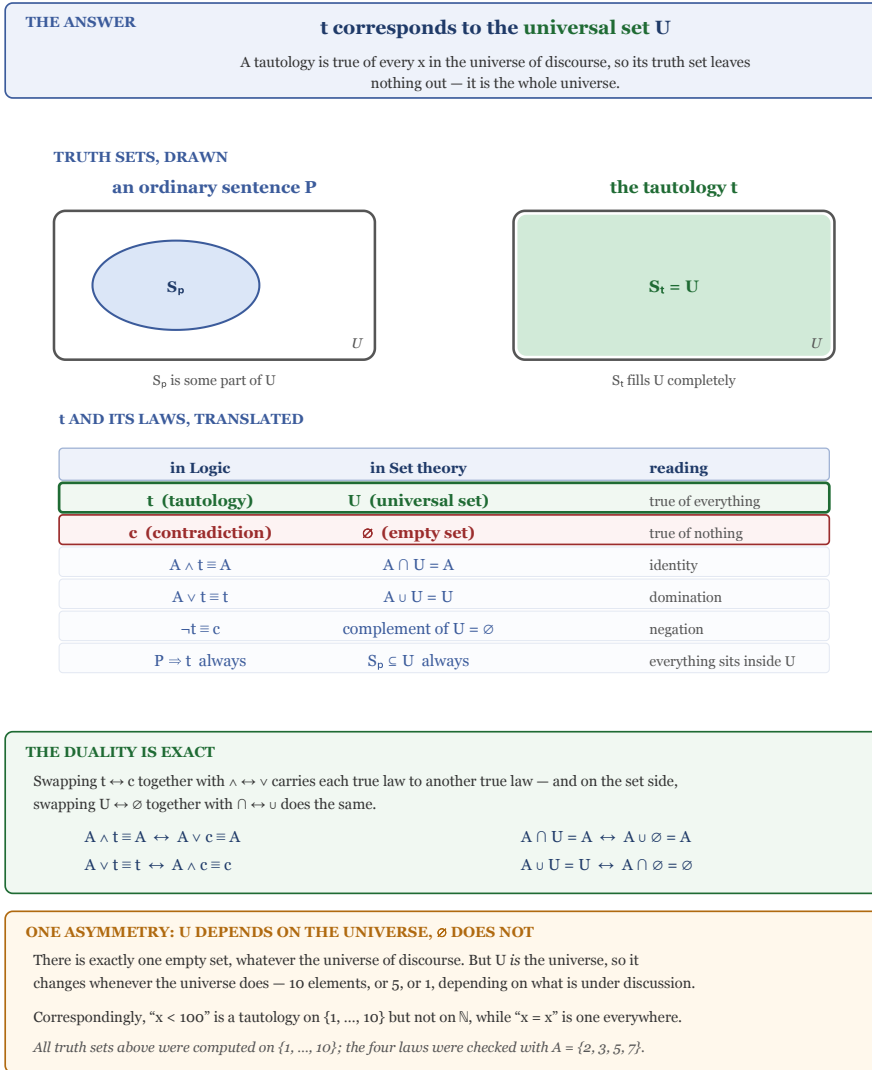


Figure 1.1: A diagram in four parts. The first gives the answer: t corresponds to the universal set U . A tautology is true of every x in the universe of discourse, so its truth set leaves nothing out and is the whole universe. The second part draws two rectangles each labelled U in the corner. On the left, in blue, an ordinary sentence P has a small oval truth set S_P sitting inside the rectangle, with most of the rectangle left white; the caption reads S_P is some part of U . On the right, in green, the tautology t has its truth set shaded across the entire rectangle, labelled $S_t = U$, with the caption S_t fills U completely. A table then translates t and its laws. The tautology t corresponds to the universal set U , read as true of everything, and this row is highlighted. The contradiction c corresponds to the empty set, read as true of nothing. A and t is equivalent to A , corresponding to $A \cap U = A$, an identity law. A or t is equivalent to t , corresponding to $A \cup U = U$, a domination law. $\neg t$ is equivalent to c , corresponding to the complement of U being the empty set. And P implies t always, corresponding to the truth set of P always being a subset of U , since

everything sits inside U . A band below states that the duality is exact: swapping t with c together with and with or carries each true law to another true law, and on the set side swapping U with the empty set together with intersection and union does the same. Four pairings are shown: A and t equals A pairs with A or c equals A , matching A intersect U equals A paired with A union the empty set equals A ; and A or t equals t pairs with A and c equals c , matching A union U equals U paired with A intersect the empty set equals the empty set. A final band notes one asymmetry: U depends on the universe while the empty set does not. There is exactly one empty set whatever the universe of discourse, but U is the universe, so it changes whenever the universe does — 10 elements, or 5, or 1, depending on what is under discussion. Correspondingly, x less than 100 is a tautology on the set 1 through 10 but not on the natural numbers, while x equals x is one everywhere. All truth sets were computed on 1 through 10 and the four laws checked with A equal to the set 2, 3, 5, 7.

1.5 Remark — The Two Exercises Are One Duality

Problems 4 and 5 are a matched pair, and seeing them together is more useful than either alone:

Logic	Set theory	truth set
c (contradiction)	$\emptyset$	nothing satisfies it
t (tautology)	U	everything satisfies it

The logical operations $\wedge$ and $\vee$ are themselves dual, and the duality is **exact**: interchanging $t \leftrightarrow c$ together with $\wedge \leftrightarrow \vee$ carries any true law to another true law. On the set side the same swap reads $U \leftrightarrow \emptyset$ with $\cap \leftrightarrow \cup$.

$$\begin{array}{c|c|c|c} A \wedge t \cong A & A \vee c \cong A & A \cap U = A & A \cup \emptyset = A \\ A \vee t \cong t & A \wedge c \cong c & A \cup U = U & A \cap \emptyset = \emptyset \end{array}$$

All four set identities verified on $\{1, \dots, 10\}$ with $A = \{2, 3, 5, 7\}$. Reading across each row: the same fact, said four ways.

This is why one only needs to remember half of the laws in either subject. Given $A \cup \emptyset = A$, the dual $A \cap U = A$ follows by symbol-swapping, with no separate argument.

1.6 Remark — U Is Where Complementation Lives

The universal set is not merely a curiosity at the top of the lattice; it is what makes **complement** a well-defined operation.

The logical negation $\neg P$ is perfectly meaningful on its own. Its set counterpart, though, requires a reference:

$$S_{\neg P} = U \setminus S_P,$$

the elements of the universe *not* in S_P . Without a fixed U there is no such thing as “the complement of A ” — one can only speak of $B \setminus A$ relative to some chosen B .

This is exactly the role t plays in the logical identities: $\neg t \cong c$ becomes “the complement of U is $\emptyset$ ”, and $A \vee \neg A \cong t$ becomes

$$A \cup (U \setminus A) = U,$$

the statement that a set and its complement exhaust the universe. Both were checked.

1.7 Remark — One Asymmetry Between the Two Answers

The pairing is clean but not perfectly symmetric, and the difference is worth stating.

There is exactly one empty set. Whatever universe one works in, $\emptyset$ is the same object — extensionality forces it, as noted in the previous problem.

There is no single universal set. U is the universe of discourse, so it changes as soon as the discussion does. Working over $\{1, \dots, 10\}$ gives a ten-element U ; over $\{1, \dots, 5\}$ a five-element one; over $\mathbb{N}$ an infinite one.

The logical side shows the same asymmetry. Whether a sentence counts as a tautology can depend on the universe: on $\{1, \dots, 10\}$ the sentence “ $x < 100$ ” has truth set U and so qualifies, but on $\mathbb{N}$ it does not. By contrast “ $x = x$ ” is a tautology in every universe. Contradictions behave similarly, but the *empty set itself* does not change — only which sentences pick it out.

1.8 Remark — Why There Is No Set of Everything

The text mentions sidestepping foundational issues here, and this exercise is where one of them surfaces.

It is tempting to want a single absolute U — “the set of all sets” — so that t would have one canonical truth set. **No such set exists.** If U contained every set, consider

$$R = \{x \in U : x \notin x\}.$$

Asking whether $R \in R$ gives a contradiction either way: if $R \in R$ then by definition $R \notin R$, and if $R \notin R$ then R qualifies for membership, so $R \in R$. This is **Russell’s paradox**, and it is why modern set theory always works within some fixed but unspecified universe rather than a universal one.

So the honest form of the answer is: t corresponds to the universal set *of the universe of discourse currently in play*. The correspondence is perfectly good — it simply comes with a context, in a way that the answer $\emptyset$ to the previous exercise does not.